

Preprocessing_data

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1 Set-up

```
#loading packages  
  
library(here)  
library(tidyverse)
```

2 Preprocessing the Spanish native data

```
#Loading Kocher's Spanish native data  
  
kocher_native_data <- read.csv(here::here("data", "raw_data", "datanatives.csv"))  
  
# Adding the 'SubjectGenus' variable to the Kocher's dataframe  
  
kocher_native_data[, 'SubjectGenus'] = NA  
  
#write_csv(kocher_native_data, here("data", "kocher_native_data.csv")) #saving added data as
```

3 Preprocessing the L1 German data

```
#Loading Kocher's L1 German data and the additionally extracted L1 German data with more spee  
  
raw_data_germanL1 <- read.csv(here::here("data", "raw_data", "raw_data_germanL1.csv"), sep = "
```

```

kocher_data_germanL1 <- read.csv(here::here("data", "kocher_germanL1_data.csv")) #Kocher's a

#Extracting the newly added speakers
new_germanL1 <- anti_join(raw_data_germanL1, kocher_data_germanL1,
                          by = "Filename") #17 in total, not segmented nor annotated yet

```

The next step consists of some data wrangling:

```

# Changing abbreviations of column 'Proficiency' to full speaker proficiency level names in t

kocher_data_germanL1 <- kocher_data_germanL1 |>
  mutate(
    Proficiency = case_when(
      Proficiency == "LB" ~ "Lower beginner",
      Proficiency == "UB" ~ "Upper beginner",
      Proficiency == "LI" ~ "Lower intermediate",
      Proficiency == "UI" ~ "Upper intermediate",
      Proficiency == "LA" ~ "Lower advanced",
      Proficiency == "UA" ~ "Upper advanced",
      TRUE ~ Proficiency)
  )

```

Further, I need to change the column names of the dataframe 'new_germanL1' so that they agree with those of the dataframe 'kocher_data_germanL1' and deselect irrelevant columns in 'new_germanL1' df. After that, I add the language proficiency level in numbers (level 1-6), in addition to the different levels names.

```

colnames(new_germanL1) #current column names

```

```

[1] "Subcorpus"
[2] "Filename"
[3] "Year.data.collection"
[4] "Placement.test.score..raw."
[5] "Placement.test.score...."
[6] "Proficiency"
[7] "Sex"
[8] "Age"
[9] "School.University.Institution"
[10] "Major"
[11] "Year.at.university.school"
[12] "L1"

```

```

[13] "Father.s.native.language"
[14] "Mother.s.nativelanguage"
[15] "Languages.spoken.at.home"
[16] "Age.of.exposure.to.Spanish"
[17] "Years.studying.Spanish"
[18] "Stay.abroad.in.Spanish.speaking.country.....1.month."
[19] "Stay.abroad..where."
[20] "Stay.abroad..when."
[21] "Stay.abroad..months."
[22] "Language.certificates..type.and.level."
[23] "Proficiency..self.assessment..speaking"
[24] "Proficiency..self.assessment..listening"
[25] "Proficiency..self.assessment..reading"
[26] "Proficiency..self.assessment..writing"
[27] "Proficiency..self.assessment."
[28] "Additional.foreign.language.s."
[29] "Proficiency..self.assessment..in.additional.language.speaking"
[30] "Proficiency..self.assessment..in.additional.language.listening"
[31] "Proficiency..self.assessment..in.additional.language.reading"
[32] "Proficiency..self.assessment..in.additional.language.writing"
[33] "Medium"
[34] "Task.number"
[35] "Task.title"
[36] "Writting.audio.details"
[37] "Minutes.taken.to.complete.the.task"
[38] "Where.the.task.was.done"
[39] "Resources.used"
[40] "Text"
[41] "Original.text"

```

```

new_germanL1 <- new_germanL1 |> #renaming columns as in Kochers df
  rename(
    "Year" = "Year.data.collection",
    "AgeExposureSpanish" = "Age.of.exposure.to.Spanish",
    "YearsstudySpanish" = "Years.studying.Spanish",
    "AbroadSpanishCountry" = "Stay.abroad.in.Spanish.speaking.country.....1.month.",
    "DurationTask" = "Minutes.taken.to.complete.the.task",
    "LocationTask" = "Where.the.task.was.done",
    "ForeignLanguage" = "Additional.foreign.language.s.",
    "Score" = "Placement.test.score...",
    "Place" = "School.University.Institution",
    "FullText" = "Text"
  )

```

```

) |>
select( #deselecting columns not needed
  -Subcorpus, -Placement.test.score..raw., -L1, -Major, -Year.at.university.school, -Father
mutate( #adding ProficiencyNr column and values 1-6
  ProficiencyNr = case_when(
    Proficiency == "Lower beginner" ~ "1",
    Proficiency == "Upper beginner" ~ "2",
    Proficiency == "Lower intermediate" ~ "3",
    Proficiency == "Upper intermediate" ~ "4",
    Proficiency == "Lower advanced" ~ "5",
    Proficiency == "Upper advanced" ~ "6",
    TRUE ~ Proficiency)
)

```

Next, I want to add the 'NSLhome' and 'NSLadditional' columns and therefore extract the information of the column 'Languages.spoken.at.home'.

```
table(new_germanL1$Languages.spoken.at.home) # displaying languages spoken at home
```

German	German, English	German, Italian	German, Serbian
10	1	1	1
Spanish	Spanish, English	Spanish, German	Turkish, Spanish
1	1	1	1

#Apart from German, some speakers also grew up acquiring English, Italian, Serbian, Spanish a

```
class(new_germanL1$Languages.spoken.at.home) #vector type "character"
```

```
[1] "character"
```

```
nsl_home_L1germ <- c("Spanish", "Italian", "Serbian", "Turkish")
```

```

new_germanL1 <- new_germanL1 |>
mutate(
  NSLhome = case_when(
    str_detect(Languages.spoken.at.home, paste(nsl_home_L1germ, collapse = "|")) ~ "Yes",
    TRUE ~ "No"
  )
) #If the speakers grew up with a NSL at home the answer is "Yes", else "No"

```

```
#Adding the NSLadditional column
```

```
class(new_germanL1$ForeignLanguage) #vector type "character"
```

```
[1] "character"
```

```
table(new_germanL1$ForeignLanguage) # displaying languages spoken at home
```

```
English Italian  
      16      1
```

```
new_germanL1 <- new_germanL1 |>  
  mutate(NSLadditional = case_when(  
    NSLhome == "Yes" ~ "Yes",  
    NSLhome == "No" & ForeignLanguage == "Italien" ~ "Yes",  
    TRUE ~ "No"  
  )) #When speakers already acquired a NSL at home than column "NSLadditional" will already l
```

Before merging the two dataframes I need the values in the column 'ProficiencyNr' to be the same vector type:

```
class(kocher_data_germanL1$ProficiencyNr) #integer
```

```
[1] "integer"
```

```
class(new_germanL1$ProficiencyNr) #character
```

```
[1] "character"
```

```
new_germanL1 <- new_germanL1 |>  
  mutate(ProficiencyNr = as.integer(ProficiencyNr)) #integer now
```

```
#write_csv(new_germanL1, here("data", "new_germanL1.csv")) #saving added data as .csv file
```

Now I can segment each clause per row of every text. Therefore I use the Large Language Models (LLMs) *Meta Llama 3.3 70B* and *DeepSeek R1 Distill LLama 70B* Instruct for this process via the University of Cologne's academic cloud access. As this corpus contains learner data most of the clauses were not segmented correctly which is why I had to manually recognise and segment the remaining clauses. The `new_germanL1_segmented` dataframe is then read into R again for merging.

```
#Reading the manually segmented file in R

new_germanL1_segmented <- read.csv(here::here("data","new_germanL1_segmented.csv"), stringsAsFactors = FALSE)

# Combining the segmented clauses of the dataframe 'new_germanL1_segmented' with the metadata of the 'new_germanL1' dataframe
joined_new_germanL1_segmented <- inner_join(x = new_germanL1_segmented, y = new_germanL1, by = "ID")

#I obtain a dataframe with 18 variables/columns as 'FullText' was added to the remaining 17 variables
```

Now, I merge the dataframe 'joined_new_germanL1_segmented' containing only the clauses, filenames and metadata of the new added L1German data to Kocher (2025b)'s already annotated clauses using the function `bind_rows()`. The new dataframe's name is *joined_full_germanL1*.

```
joined_germanL1_data <- bind_rows(kocher_data_germanL1, joined_new_germanL1_segmented)

#Afterwards the column 'Thetarole' was deselected as I consider this to be a very subjective variable

joined_germanL1_data <- joined_germanL1_data |>
  select(-Thetarole, -SentenceType)

#Adding the 'SubjectGenus' variable to the 'joined_germanL1_data' dataset

joined_germanL1_data[, 'SubjectGenus'] = NA

#write_csv(joined_germanL1_data, here("data", "joined_germanL1_data.csv")) #saving added data to csv file
```

This .csv file can now be manually annotated. Concretely, I annotate the values of the columns:

- 'Textnr'
- 'Sentence'
- 'ID'
- 'SubjectPosition'

- ‘ClauseType’
- ‘Specificity’
- ‘InformationStructure’
- ‘ReferentialContinuity’
- ‘Subjecttype’
- ‘Verb’
- ‘VerbConstruction’
- ‘SubjectGenus’

4 L1 German data extraction for checking on the robustness of Kocher’s annotation

For a later robustness check I will randomly extract 3 speakers of Kocher (2025b)’s dataset and re-annotate their clauses in order to check if the annotation processes align.

```
#Letting R select three Filenames randomly

set.seed(8) #Setting a seed to ensure that the outcome of my randomly generated number series is reproducible
random_filenames <- sample(kocher_data_germanL1$Filename, 3)

#[1] "DE_WR_41_23_5_14_PN"
#[2] "DE_WR_39_46_30_14_IK"
#[3] "DE_WR_41_21_3_14_MP"

#Creating a new dataframe with only the rows of the three selected filenames

sample_kocher_germanL1 <- kocher_data_germanL1 |>
  filter(Filename %in% c(
    "DE_WR_41_23_5_14_PN",
    "DE_WR_39_46_30_14_IK",
    "DE_WR_41_21_3_14_MP"))
  )

# Adding the 'SubjectGenus' column

sample_kocher_germanL1[, 'SubjectGenus'] = NA
```

```
# Deleting 'Thetarole' and 'SentenceType' columns

sample_kocher_germanL1 <- sample_kocher_germanL1 |>
  select(-Thetarole, -SentenceType) #deselecting columns not needed

#write_csv(sample_kocher_germanL1, here("data", "sample_kocher_germanL1.csv")) #saving data
```

The columns of df “sample_kocher_germanL1” will be emptied and reannotated manually.

5 Preprocessing the L1 English data

```
englishL1_data <- read.csv(here::here("data", "raw_data/raw_data_englishL1.csv"), sep = "\t")

# Changing the column names so they align with the L1German data.

colnames(englishL1_data) #current column names
```

```
[1] "Subcorpus"
[2] "Filename"
[3] "Year.data.collection"
[4] "Placement.test.score..raw."
[5] "Placement.test.score...."
[6] "Proficiency"
[7] "Sex"
[8] "Age"
[9] "School.University.Institution"
[10] "Major"
[11] "Year.at.university.school"
[12] "L1"
[13] "Father.s.native.language"
[14] "Mother.s.nativelanguage"
[15] "Languages.spoken.at.home"
[16] "Age.of.exposure.to.Spanish"
[17] "Years.studying.Spanish"
[18] "Stay.abroad.in.Spanish.speaking.country.....1.month."
[19] "Stay.abroad..where."
[20] "Stay.abroad..when."
[21] "Stay.abroad..months."
```

```

[22] "Language.certificates..type.and.level."
[23] "Proficiency..self.assessment..speaking"
[24] "Proficiency..self.assessment..listening"
[25] "Proficiency..self.assessment..reading"
[26] "Proficiency..self.assessment..writing"
[27] "Proficiency..self.assessment."
[28] "Additional.foreign.language.s."
[29] "Proficiency..self.assessment..in.additional.language.speaking"
[30] "Proficiency..self.assessment..in.additional.language.listening"
[31] "Proficiency..self.assessment..in.additional.language.reading"
[32] "Proficiency..self.assessment..in.additional.language.writing"
[33] "Medium"
[34] "Task.number"
[35] "Task.title"
[36] "Writting.audio.details"
[37] "Minutes.taken.to.complete.the.task"
[38] "Where.the.task.was.done"
[39] "Resources.used"
[40] "Text"
[41] "Original.text"

```

```

englishL1_data <- englishL1_data |> #renaming columns as in Kochers df
  rename(
    "Year" = "Year.data.collection",
    "AgeExposureSpanish" = "Age.of.exposure.to.Spanish",
    "YearsstudySpanish" = "Years.studying.Spanish",
    "AbroadSpanishCountry" = "Stay.abroad.in.Spanish.speaking.country....1.month.",
    "DurationTask" = "Minutes.taken.to.complete.the.task",
    "LocationTask" = "Where.the.task.was.done",
    "ForeignLanguage" = "Additional.foreign.language.s.",
    "Score" = "Placement.test.score....",
    "Place" = "School.University.Institution",
    "FullText" = "Text"
  ) |>
  select( #deselecting columns not needed
    -Subcorpus, -Placement.test.score.raw., -L1, -Major, -Year.at.university.school, -Father
  ) |>
  mutate( #adding ProficiencyNr column and values 1-6
    ProficiencyNr = case_when(
      Proficiency == "Lower beginner" ~ "1",
      Proficiency == "Upper beginner" ~ "2",
      Proficiency == "Lower intermediate" ~ "3",
      Proficiency == "Upper intermediate" ~ "4",

```

```

    Proficiency == "Lower advanced" ~ "5",
    Proficiency == "Upper advanced" ~ "6",
    TRUE ~ Proficiency)
)

```

Next, I add the columns 'NSLhome' and 'NSLadditional':

```
table(englishL1_data$Languages.spoken.at.home) # displaying languages spoken at home
```

English	English, Farsi, Kurdish	English, French
147	1	1
English, Gujarati, Hindi	English, Japanese	English, Spanish
1	2	2
Hindi, Gujarati	Italian, English	Portuguese
1	1	1
Punjabi	Spanish	Spanish, English
1	2	1

```
#Apart from English, some speakers also grew up acquiring Farsi, Kurdish, French, Gujarati, I
```

```
nsL_home_L1engl <- c("Farsi", "Kurdish", "Gujarati", "Hindi", "Japanese", "Spanish", "Italian
```

```
class(englishL1_data$Languages.spoken.at.home) #vector type "character"
```

```
[1] "character"
```

```

englishL1_data <- englishL1_data |>
  mutate(
    NSLhome = case_when(
      str_detect(Languages.spoken.at.home, paste(nsl_home_L1engl, collapse = "|")) ~ "Yes",
      TRUE ~ "No"
    )
  ) #If the speakers grew up with a NSL at home the answer is "Yes", else "No"

#Adding the NSLadditional column

class(englishL1_data$ForeignLanguage) #vector type "character"

```

```
[1] "character"
```

```
table(englishL1_data$ForeignLanguage) # displaying foreign languages learned
```

```

123
American Sign Language
1
Arabic
2
Arabic (Egyptian Colloquial), Arabic (MSA)
1
Arabic (MSA)
1
Catalan
1
Creole
1
French
5
German
5
German, French
1
Gujarati, Hindi
1
Hindi
1
Italian
2
Italian (A1)
1
Italian, French
1
Japanese
3
Japanese, Portuguese, French
1
Kaqchikel
1
Kurdish
```

```

1
Norwegian
1
Portuguese
1
Portuguese (A1)
1
Russian
3
Swahili
2

```

```

nsl_foreignlg_L1engl <- c("Arabic", "Catalan", "Gujarati", "Hindi", "Italian",
  "Japanese", "Portuguese", "Kaqchikel", "Kurdish",
  "Russian", "Swahili")

```

```

englishL1_data <- englishL1_data |>
  mutate(NSLadditional = case_when(
    NSLhome == "Yes" ~ "Yes",
    NSLhome == "No" & str_detect(ForeignLanguage, paste(nsl_foreignlg_L1engl, collapse = "|")
      TRUE ~ "No")
  ) #When speakers already acquired a NSL at home than column "NSLadditional" will already

```

```

#Checking if the vector type of the column 'ProficiencyNr' aligns with the L1German dataset:

```

```

class(englishL1_data$ProficiencyNr) #character

```

```

[1] "character"

```

```

englishL1_data <- englishL1_data |>
  mutate(ProficiencyNr = as.integer(ProficiencyNr)) #changed to integer now
#write_csv(englishL1_data, here("data", "englishL1_data.csv")) #saving added data as .csv fi

```

I segmented each clause per row of every text using the machine-based tool *Meta Llama 3.3 70B* and *DeepSeek R1 Distill Llama 70B Instruct* for this process via the University of Cologne's academic cloud access. The `englishL1.csv_segmented` df is then read into R again for merging.

```

#Reading the segmented file into R

englishL1_segmented <- read.csv(here::here("data", "englishL1_segmented.csv"), stringsAsFactors = FALSE)

colnames(englishL1_segmented)

[1] "Text"      "Filename"

colnames(englishL1_data)

[1] "Filename"      "Year"
[3] "Score"         "Proficiency"
[5] "Sex"           "Age"
[7] "Place"         "Languages.spoken.at.home"
[9] "AgeExposureSpanish" "YearsstudySpanish"
[11] "AbroadSpanishCountry" "ForeignLanguage"
[13] "DurationTask"      "LocationTask"
[15] "FullText"          "ProficiencyNr"
[17] "NSLhome"           "NSLadditional"

# Combining the segmented clauses of the dataframes 'englishL1_segmented' with the metadata of 'englishL1_data'
joined_englishL1_data <- inner_join(x = englishL1_segmented, y = englishL1_data, by = "Filename")

# Adding the 'SubjectGenus' variable to the 'joined_englishL1_data' dataframe for manual annotation
joined_englishL1_data[, 'SubjectGenus'] = NA
joined_englishL1_data[, 'Textnr'] = NA
joined_englishL1_data[, 'Sentence'] = NA
joined_englishL1_data[, 'ID'] = NA
joined_englishL1_data[, 'SubjectPosition'] = NA
joined_englishL1_data[, 'ClauseType'] = NA
joined_englishL1_data[, 'Specificity'] = NA
joined_englishL1_data[, 'InformationStructure'] = NA
joined_englishL1_data[, 'ReferentialContinuity'] = NA
joined_englishL1_data[, 'Subjecttype'] = NA
joined_englishL1_data[, 'Verb'] = NA
joined_englishL1_data[, 'VerbConstruction'] = NA

```

As the L1German data - including the newly added speaker texts - contain 2997 clauses, I aim to extract a similarly high number of L1English clauses that I annotate manually. Therefore, I select a random sample of 2000 clauses in the following code lines:

```

joined_englishL1_data$sample <- NA #creating sample column
set.seed(42) #Setting a seed to ensure that the outcome of my randomly generated number ser
random_clauses <- sample(nrow(joined_englishL1_data), 2000)

#Assigning the word 'sample' to the selected rows

"sample" -> joined_englishL1_data$sample[random_clauses]

#write_csv(joined_englishL1_data, here("data", "joined_englishL1_data.csv")) #saving added d

```

PRE-ANNOTATION QUESTIONS

- When L1 English speakers used English words, especially verbs as a lack of vocabulary knowledge, should I still annotate as if it was the Spanish word (e.g. *y Charlie gives un hombre él bebe* (EN_WR_24_16_0.5_14_SZG))? I haven't discovered this issue in the L1 German data so far to be able to compare it with Kocher's results.

-> I will exclude these cases containing English verbs.

- The variable NSLadditional in Kocher's L1 German speaker dataset is unclear to me. For example, the speaker 'DE_WR_27_19_6_14_AOK' grew up speaking Russian and German (which is why NSLhome is set to 'Yes' for Russian), but apart from Spanish, they only speak English. Nevertheless, Kocher assigned the value 'Yes' to NSLadditional, even though English is not an NSL and there are no other languages studied that might be an NSL. Another example: The speaker 'DE_WR_38_21_7_14_CW' learnt Hebrew alongside Spanish and did not mention English at all. Nevertheless, Kocher assigned 'English' to the ForeignLanguage column and added 'Yes' to NSLadditional. I don't understand where the languages mentioned in the 'ForeignLanguages' column or the 'NSLadditional' categorisation as 'Yes/No' are coming from. Another example regarding speaker 'DE_WR_38_21_10_14_HYR': There, Kocher classified 'Hebrew' as 'ForeignLanguage' and 'NSLadditional' as 'No', even though the person studied English and never mentions Hebrew in the raw data that I extracted. Could the metadata be incorrect?

-> In some cases the data is incorrect. But: "NSLadditional" is a general variable that is classified as "yes", when the speaker acquired a NSL at home or studied it afterwards. So when the column "NSLhome" is set to "yes", "NSLadditional" will always be "yes" too.

- Further, I annotated the column 'Textnr' not only in the new L1 German data but re-annotated it also in Kocher's pre-annotated L1 data (both merged in dataframe joined_germanL1_data) as it was partially incorrect.

ANNOTATION SCHEME:

Variable	Annotated values
SubjectPosition	PRE, POST, NA (in nullsub cases)
ClauseType	MC, SC
Specificity	DEF, INFED, NA (when nullsub is used)
Information-Structure	GI(VEN), NE(W), NA (unknown subject form)
Referential-Continuity	Y(ES), N(O), NA (unknown subject form)
Subject-Type	nullsub, dp, pron
Verb	Verb of the corresponding sentence
Verb-Construction	simple-finite, periphrasis-lexical, periphrasis-gerund, modal, infinitive
Subject-Genus	masc, fem, unknown¹, general ¹Some of the <i>unknown</i> cases, that Kocher actually claims to have excluded, are still within the annotated data. <i>Unknown</i> includes cases in which the subject is not clearly identifiable from the prior context, where an impersonal form has been used. (<i>General</i> refers to first person plural forms as in <i>Here we can see [...]</i>.) > SOLCHE FÄLLE GANZ RAUS? <i>Unknown</i> examples: • <i>cuando de repente le tiran como basura</i> from kocher_native_data 1_5; • <i>Se le ha caído algo</i> from kocher_native_data 2_10; • <i>que han de cuidarlo y darle mimos</i> from kocher_native_data 7_25; • <i>y alguien vuelve a tirar basura muy cerca de él</i> from kocher_native_data 10_7
	General

ANNOTATIONSSCHEMA:

Variable	Annotierte Werte
SubjectPosition	PRE, POST, NA (bei nullsubj)
ClauseType	MC, SC
Specificity	DEF, INFED, NA (bei nullsub)
InformationStructure	GI(VEN), NE(W), NA (bei unbekannter Subjektform)
ReferentialContinuity	Y(ES), N(O), NA (bei unbekannter Subjektform)
SubjectType	nullsub, dp, pron
Verb	Verb des jeweiligen Clauses
VerbConstruction	simple-finite, periphrasis-lexical, periphrasis-gerund, modal, infinitive
SubjectGenus	masc, fem, unknown, general

Some notes to explain the variables in Kocher (2025b)’s and my data annotation:

- The ‘Foreign language’ column seems to refer only to additional foreign languages studied, rather than to other L1s acquired at home (NSLhome). Kocher (2025b) simply categorised whether speakers had acquired a null subject language at home, without specifying which one. In this case, ‘No’ could mean that the speaker has not acquired another language at all, or that the language they grew up with was not a null subject language.
- Kocher (2025b) separated the subordinate clauses (SCs), such as ‘Mientras la gente [que vive en los edificios] tiran su basura de la ventana a la calle’, from the main clause (MC) and displayed them in the row below.
- Cut-out SCs were marked as [...] within the MCs.
- Direct speech was also marked in brackets and not annotated. Lines containing English direct speech were omitted (e.g. kocher_data_germanL1 ID9_28), but not when the direct speech was translated into Spanish (e.g. kocher_data_germanL1 ID59_9). The annotation was inconsistent: for example, in line 2328, direct speech was marked with ‘[...]’, as I did too.
- Further, I excluded meta commentaries in the L1 English dataset, such as *La película es en negro y blanco, pero es muy bueno* (in EN_WR_14_23_1_14_BB), as well as comments written in the first person singular or plural, such as *i do not know what else to write i already did my best with the previous sentences i wrote* (EN_WR_20_21_4_14_PKO) or *Al inizio del short lo vemos vestido mal con la ropa desarreglada* (EN_WR_39_62_2_14_MA) before the annotation process, in order to avoid their accidental inclusion in the random sample. I decided to exclude them, as there are too few first-person sentences to obtain reliable results about these subjects’ behaviour and make meaningful model interpretations. In contrast, Kocher (2025b) mostly deleted them after assigning an ID number, as some ID numbers are missing

from the final annotated document (see the `kocher_germanL1_data` file). However, some clauses containing first-person singular or plural subjects remained in her annotation and consequently influenced the model (see e.g. *i creó* in `joined_germanL1_data_annotation` (DE_WR_31_27_24_14_MS no. 19_2), in `joined_germanL1_data_annotation` (*Nunca aprendí español* DE_WR_20_50_2_14_HO no. 4_2).

- I took over the same exclusion criteria as Kocher (2025a) which applies to imperatives, indefinite “se” constructions and impersonal sentences (e.g., *hay*, *llueve*). Also I decided to exclude sentences that do not contain a verb (see *una película sobre un bebé encontrado y buscando amor* EN_WR_12_30_0.7_14_NC no. 6_1) or in which the student used an English word (see *La cop chase the el hombre* EN_WR_12_30_0.7_14_NC no. 6_5). These sentences were not deleted but taking off the *sample* selection in order to still be able to follow the referential continuity and information structure of the subsequent sentences.

- Kocher, A. (2025a). The subject realization in L2 Spanish by German L1 speakers: A corpus study. *Isogloss. Open Journal of Romance Linguistics*, 11(2), 1–34. <https://doi.org/10.5565/rev/isogloss.437>
- Kocher, A. (2025b). The subject realization in L2 spanish by german L1 speakers: A corpus study. *Isogloss. Open Journal of Romance Linguistics*, 11(2), 1–34. <https://doi.org/10.5565/rev/isogloss.437>